

Fengyuan Yang

Video Generation / Generative World Models
/ Human-centric 3D/4D Perception

☎ +65 89420532 / +86 15810536858

✉ fengyuan.yang@u.nus.edu

🌐 martayang.github.io

🐙 GitHub



Research Profile

Ph.D. candidate at NUS working on controllable video generation, generative world modeling, and human-centric 3D/4D vision. My research decouples human motion, camera trajectories, and scenes from video, then uses these factors as independently editable controls to recompose coherent videos.

Education

- 08.2022 – Present **National University of Singapore (NUS)**
Ph.D. in Computer Science, Department of Computer Science, Singapore
Advisor: [Prof. Angela Yao](#) | Expected early 2027
- 09.2019 – 06.2022 **Institute of Computing Technology (ICT), Chinese Academy of Sciences**
Master's in Computer Science, Beijing, China | GPA: **3.92/4.00**
Advisors: [Prof. Xilin Chen](#) and [Prof. Ruiping Wang](#)
- 02.2018 – 07.2018 **École Polytechnique Fédérale de Lausanne (EPFL)**
Exchange in Computer Science, Lausanne, Switzerland | GPA: **5.71/6.00**
- 09.2015 – 07.2019 **University of Chinese Academy of Sciences (UCAS)**
B.Eng. in Computer Science and Technology, Beijing, China | GPA: **3.89/4.00**

Research and Internship Experience

- 07.2025 – Present **Research Intern, Baidu AMU**
Compositional video generation | Mentor: [Dr. Hang Zhou](#) | [Code](#) | [Project Page](#)
- Developed **ONE-SHOT** on Wan2.1-14B to generate long, coherent videos with plausible interactions from decoupled identity, motion, scene, and camera controls.
 - Designed spatial RoPE cross-attention to inject human motion, with LoRA-adapted conditioning and context memory for minute-long identity and scene consistency.
 - Conducted end-to-end research from data preparation and model training to evaluation, resulting in a first-author paper at **NeurIPS 2026**.
- 08.2022 – Present **Ph.D. Research, National University of Singapore**
- Developed **KITRO** for 2D-guided, kinematic-tree-aware mesh refinement, **Humans-as-Checkerboards** for human-guided monocular SLAM scale calibration, and **HumanBA** to jointly refine global human motion and camera trajectories.
 - These works progress from 2D-guided human reconstruction to joint 4D reconstruction of people, scenes, and cameras, yielding three first-author papers at **CVPR 2024, ICCV 2025, CVPR 2026**.
- 09.2019 – 06.2022 **Master's Research, Institute of Computing Technology**
Semantic-guided few-shot learning: integrated semantic knowledge with visual prototypes and local features, yielding two first-author papers at **WACV 2022, WACV 2023**.

Technical Skills

Programming	Python, C/C++; PyTorch, Diffusers.
Generative Models	Diffusion / Flow Matching; Transformers (DiT); Multimodal Conditioning.
Training	LoRA Fine-tuning; DeepSpeed; Multi-GPU Training; Data Processing and Evaluation.
3D/4D Vision	Human Mesh Recovery (SMPL/SMPL-X); SLAM; Bundle Adjustment.
Scene Geometry	Feed-forward 3D Reconstruction; Camera Geometry and Pose Estimation.

First-Author Publications

- NeurIPS 2026** **Fengyuan Yang** et al. “**ONE-SHOT**: Compositional Human-Environment Video Synthesis via Spatial-Decoupled Motion Injection and Hybrid Context Integration.”
Video generation: freely compose independent identity, motion, scene and camera.
- CVPR 2026** **Fengyuan Yang** et al. “**HumanBA**: Human-Aware Bundle Adjustment via Global Human-Camera Decoupling.”
World-coordinate 4D reconstruction: human cues refine camera and body estimates.
- ICCV 2025** **Fengyuan Yang** et al. “**Humans as Checkerboards**: Calibrating Camera Motion Scale for World-Coordinate Human Mesh Recovery.”
World-coordinate recovery: human scale aligns human motion, 3D scenes and camera paths.
- CVPR 2024** **Fengyuan Yang** et al. “**KITRO**: Refining Human Mesh by 2D Clues and Kinematic-tree Rotation.”
3D human mesh refinement: 2D-guided rotations align reconstructed poses with images.
- WACV 2023** **Fengyuan Yang** et al. “Semantic Guided Latent Parts Embedding for Few-Shot Learning.”
Few-shot recognition: use semantic-guided local representations to distinguish novel classes.
- WACV 2022** **Fengyuan Yang** et al. “**SEGA**: Semantic Guided Attention on Visual Prototype for Few-Shot Learning.”
Few-shot recognition: improve visual class prototypes with semantic attention.

Honors and Awards

Research

- 09.2024 **ECCV 2024 Outstanding Reviewer Award**
- 10.2023 **SoC Research Incentive Award**, National University of Singapore

Competitions

- 06.2020 CVPR 2020 CLVision Workshop Challenge, **1st place, NI track**
- 09.2018 National Student Computer System Capability Challenge, **Grand Prize**
- 12.2017 ACM-ICPC Asia Regional Contest EC-Final, **Bronze Medal**
- 10.2017 ACM-ICPC Asia Regional Contest Xi'an, **Bronze Medal**
- 10.2016 China Collegiate Programming Contest (CCPC), Hefei, **Bronze Medal**

Academic Honors

- 07.2019 **Outstanding Undergraduate Award**, University of Chinese Academy of Sciences
- 07.2019 **Outstanding Undergraduate Award**, Beijing Municipal Education Commission

Additional Projects

- 06.2020 – 06.2022 **Robot Vision Exploration, VIPL research group**
Designed and maintained a customized ROS communication framework for LoCoBot.
- 10.2019 – 06.2022 **Visual Knowledge Base, VIPL research group**
Developed data annotation and collection guidelines for knowledge graph construction.
- 10.2018 – 10.2019 **Online Learning Collaboration, VIPL research group and Huawei**
Contributed a few-shot learning module using semantic-guided visual features.

杨丰远

视频生成 / 生成式世界模型 / 人体 3D/4D 感知

+65 89420532 / +86 15810536858

fengyuan.yang@u.nus.edu

martayang.github.io

GitHub



研究概述

新加坡国立大学计算机科学博士生，研究可控视频生成、生成式世界模型、以人为中心的三维和四维视觉。从视频中解耦并重建人体运动、相机轨迹与场景，进一步将这些因素作为可独立编辑的控制条件，生成时空一致且交互合理的视频。

教育背景

2022.08 – 至今

新加坡国立大学 (NUS)

计算机科学博士 计算机科学系，新加坡 | 预计 2027 年初毕业

导师: Angela Yao 教授

2019.09 – 2022.06

中国科学院计算技术研究所 (ICT)

计算机科学硕士 北京，中国 | GPA: 3.92/4.00

导师: 陈熙霖教授、王瑞平教授

2018.02 – 2018.07

瑞士洛桑联邦理工学院 (EPFL)

计算机科学交换生 洛桑，瑞士 | GPA: 5.71/6.00

2015.09 – 2019.07

中国科学院大学 (UCAS)

计算机科学与技术工学学士 北京，中国 | GPA: 3.89/4.00

研究与实习经历

2025.07 – 至今

研究实习生，百度 AMU

组合式可控视频生成 | Mentor: 周航博士 | [项目代码](#) | [项目主页](#)

- 基于 Wan2.1-14B 开发 **ONE-SHOT**，以人物身份、动作、场景与相机轨迹为独立解耦的控制条件，生成长程一致、交互合理的视频。
- 设计空间 RoPE 对齐的交叉注意力以注入人体运动条件，通过 LoRA 适配条件分支，并引入上下文记忆，保持分钟级视频的人物与场景一致性。
- 负责数据准备、模型训练与评测全流程研究，一作论文发表于 **NeurIPS 2026**。

2022.08 – 至今

博士研究课题，新加坡国立大学

- 研究二维关节辅助的三维人体关节树旋转修正 (**KITRO**)、基于人体尺度的单目 SLAM 尺度标定 (**Humans-as-Checkerboards**)，以及人体重建与相机轨迹估计的联合优化 (**HumanBA**)。
- 研究从二维信息辅助的人体重建拓展至人体、场景与相机的联合四维重建，期间发表三篇一作论文: **CVPR 2024**、**ICCV 2025**、**CVPR 2026**。

2019.09 – 2022.06

硕士研究课题，中国科学院计算技术研究所

将语义知识引入视觉原型与局部特征学习，提升小样本识别能力，期间发表两篇一作论文: **WACV 2022**、**WACV 2023**。

专业技能

编程与框架

Python、C/C++; PyTorch、Diffusers。

生成模型

扩散模型、Flow Matching、Transformers (DiT)、多模态条件控制。

训练与评测

LoRA 微调、DeepSpeed、多 GPU 训练、视频数据处理与模型评测。

三维与四维视觉

人体网格重建 (SMPL/SMPL-X)、SLAM、Bundle Adjustment。

场景与相机

前馈式三维重建、相机几何与位姿估计。

一作论文

- NeurIPS 2026** Fengyuan Yang et al. **"ONE-SHOT: Compositional Human-Environment Video Synthesis via Spatial-Decoupled Motion Injection and Hybrid Context Integration."**
可控视频生成：独立控制并自由组合人物身份、动作、场景与相机运动。
- CVPR 2026** Fengyuan Yang et al. **"HumanBA: Human-Aware Bundle Adjustment via Global Human-Camera Decoupling."**
世界坐标系四维重建：以人体运动约束改善相机轨迹与全局人体估计。
- ICCV 2025** Fengyuan Yang et al. **"Humans as Checkerboards: Calibrating Camera Motion Scale for World-Coordinate Human Mesh Recovery."**
统一世界坐标系重建：以人体尺度标定 SLAM，对齐人体运动、三维场景与相机轨迹。
- CVPR 2024** Fengyuan Yang et al. **"KITRO: Refining Human Mesh by 2D Clues and Kinematic-tree Rotation."**
三维人体重建优化：利用二维关节点修正关节树旋转，使重建姿态与画面中的动作更吻合。
- WACV 2023** Fengyuan Yang et al. **"Semantic Guided Latent Parts Embedding for Few-Shot Learning."**
小样本识别：利用语义引导的局部特征，提升新类别识别能力。
- WACV 2022** Fengyuan Yang et al. **"SEGA: Semantic Guided Attention on Visual Prototype for Few-Shot Learning."**
小样本识别：利用语义注意力改进视觉类别原型。

荣誉与奖项

科研认可

- 2024.09 **ECCV 2024 杰出审稿人奖**, ECCV 组委会
- 2023.10 **SoC Research Incentive Award**, 新加坡国立大学

竞赛奖项

- 2020.06 CVPR 2020 CLVision Workshop Challenge, **NI 赛道第一名**
- 2018.09 “龙芯杯”全国大学生计算机系统能力大赛, **特等奖**
- 2017.12 ACM-ICPC 亚洲区域赛 EC-Final, **铜牌**
- 2017.10 ACM-ICPC 亚洲区域赛西安站, **铜牌**
- 2016.10 中国大学生程序设计竞赛 (CCPC) 合肥站, **铜牌**

学业荣誉

- 2019.07 **优秀毕业生**, 中国科学院大学
- 2019.07 **优秀毕业生**, 北京市教育委员会

其他项目

- 2020.06 – 2022.06 **机器人视觉探索, VIPL 研究组**
设计并维护基于 ROS 的 LoCoBot 定制通信框架。
- 2019.10 – 2022.06 **视觉知识库, VIPL 研究组**
制定用于知识图谱构建的数据采集与标注规范。
- 2018.10 – 2019.10 **在线学习合作, VIPL 研究组与华为**
研发小样本学习模块，利用语义信息改进视觉特征学习。